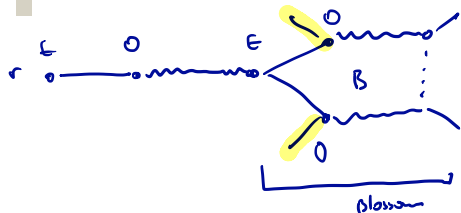


Contracting Blossoms



Need to show that contracting B preserves M -aug path.

Lemma: Let G/B denote the contracted graph, M/B denote the contracted matching.

$\exists M$ -aug path in G iff $\exists M/B$ -aug path in G/B .

" $\Leftarrow$ " Suppose $\exists$ M/B - avg path.

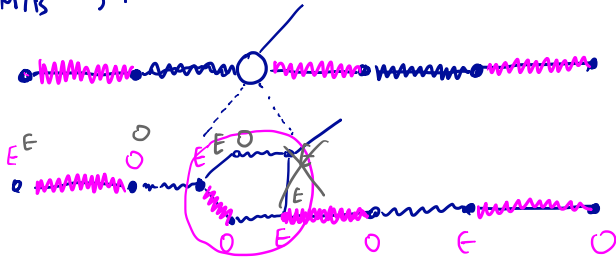
suppose any path goes through B (o/w done).

Follow M/B until we get to B .

Then we continue in one dir. or the other :

M/B - any path

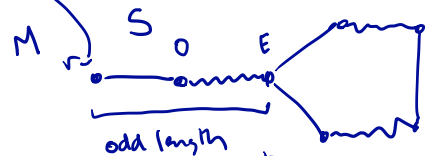
Ex:



$\swarrow G/B$

Proof of Lemma — 2/3

" $\Rightarrow$ " Equiv to show that M is
 not maximum size $\Rightarrow M/B$ is not max.



swap edges in/out
 $\downarrow$ of matching
 in stem



The exposed vertex is on B .

$|M'| = |M|$ since stem
 $\Rightarrow$ odd length. exposed

$$M' = M \Delta S$$

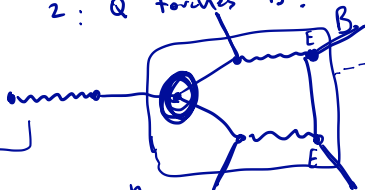
Proof of Lemma — 3/3

By assumption of M not max, M' not max

$\Rightarrow \exists M'$ -aug path: Q .

Case 1: Q avoids B . Done!

2: Q touches B .



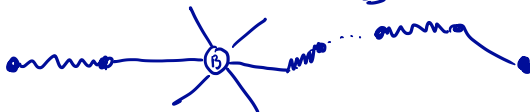
Q is aug path

Claim: Q/B is M'/B -Aug in G/B .

Indeed: in G/B , entire blossom B corresponds to a single vertex which must be exposed.

Also M'/B -aug path in G/B .

$\rightarrow$ in G/B :



~~QED~~

Edmonds's Algorithm

1. Construct maximal M -alt trees.
 - If $\nexists$ $E-E$ edge, then M is maximum.
 - If $\exists$ $E-E$ edge, contract blossom and continue.
 - If $\exists$ Aug path in $G/B_1/B_2/B_3/\dots/B_k$ then find the corresponding aug path in G .
2. If no aug path and no more $E-E$ edges, stop.

Correctness upon Termination

Thm: If alg. terminates,
then M is maximal.

pf: When terminates, current matching

in $G/B_1/\dots/B_k$ is maximum size, bec
no $E-E$ edges (sufficiency lemma).

$\Rightarrow \nexists$ any path in $G/B_1/\dots/B_k$

$\Rightarrow \nexists$ any path in G by prev lemma. $\square$

Polynomial Time Termination

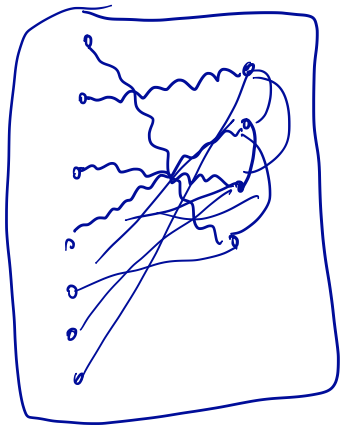
Building n -alt. trees is
poly-time.

We can contract at most $|V|$
blossoms before either terminating
with no $E-E$ edges, or before finding
an aug path.

But the matching can be augmented
at most $|V|$ times.

$$\underline{E - E}$$

If no $E-E$ edges
Even Odd

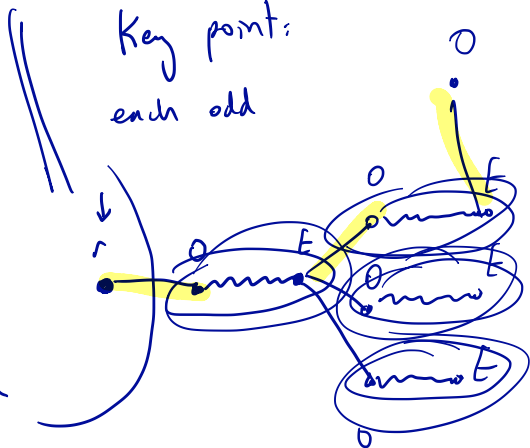


In bipartite
graph

no $O-O$ edges.

$$|E| = |O| + \underline{|X|}$$

Key point:
each odd



$$|\text{Even}| = |\text{odd}| + |X|$$

M not maximum.

